NASEM / NOVEMBER 17, 2026 · 0:20

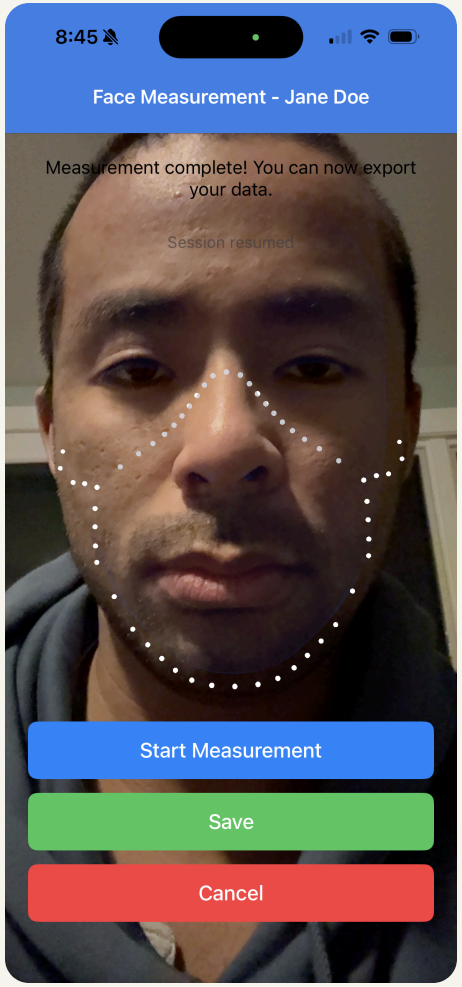
From face capture to useful measurements.

Lessons from building MasqFit and Breathesafe

Edderic Ugaddan · Breathesafe LLC

For NPPTL researchers developing iOS and Android apps

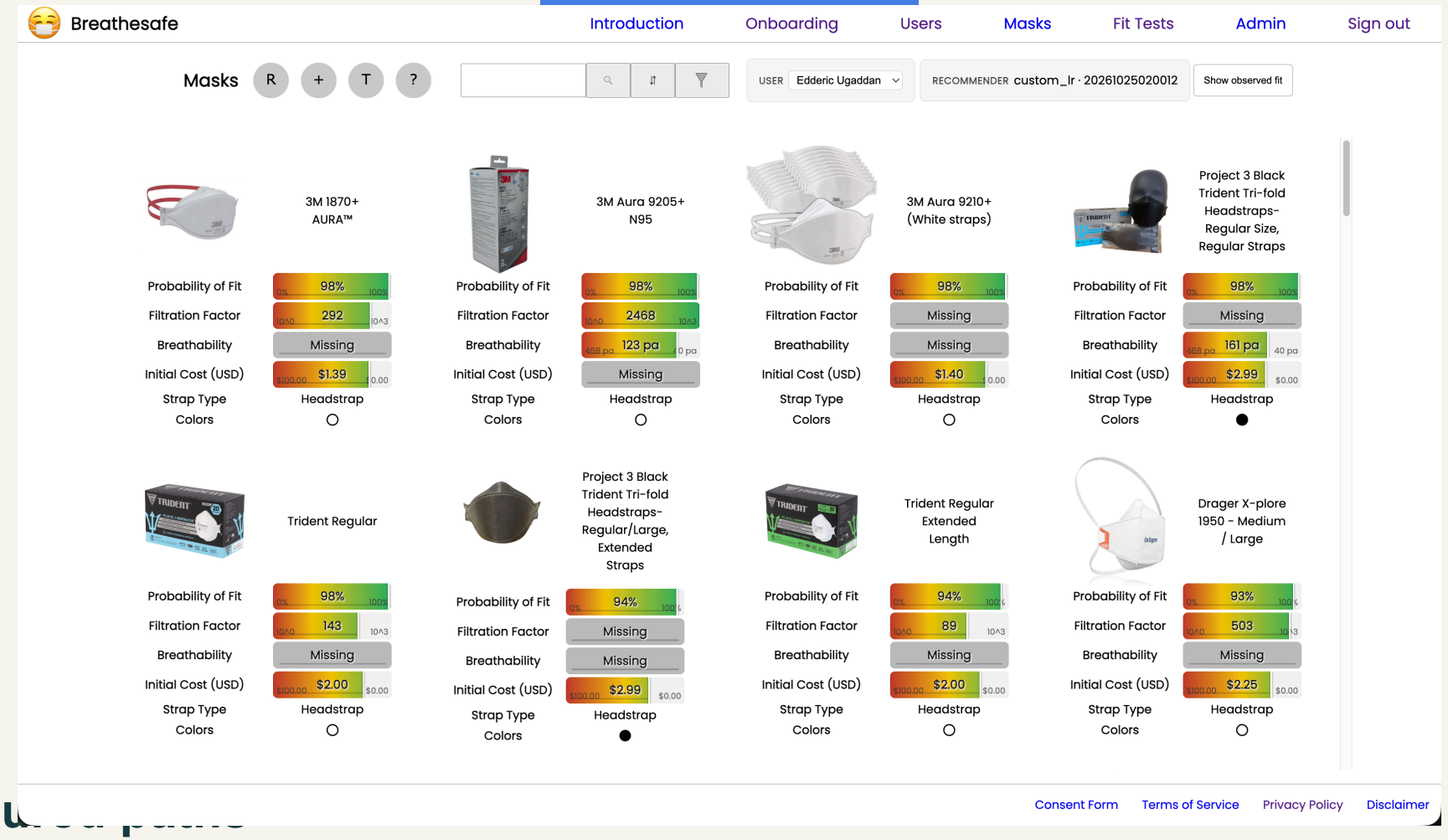
ARKit in MasqFit: five facial paths.



ARKit in the current app

What the model recommends for my face.





October 29 screenshot · October 25 custom_lr model · Observed Fit hidden

Five measured patches

Nose · Strap · Top cheek

Mid cheek · Chin

Presenter screenshot · Individual scores are model outputs; calibration is not established.

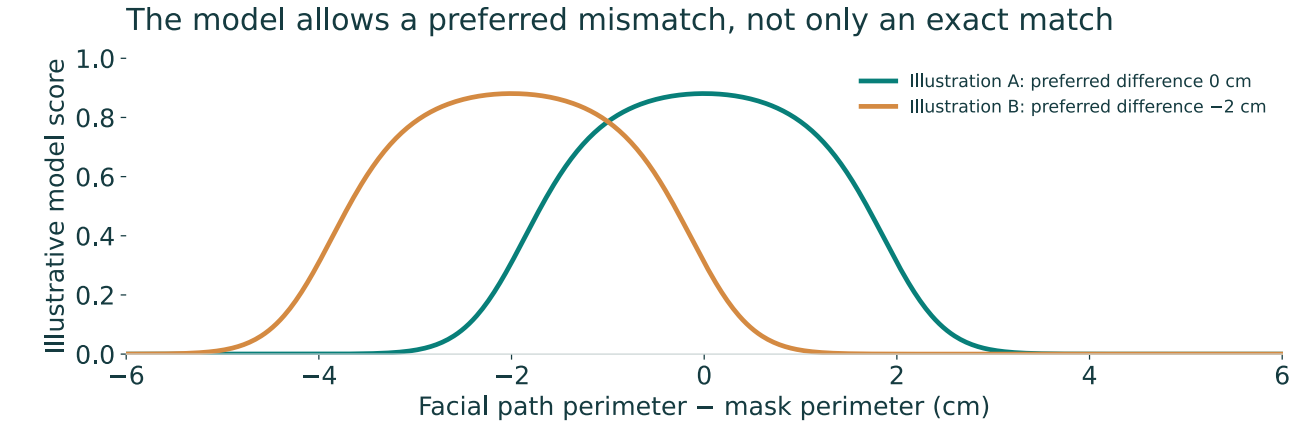
Nose, top cheek, mid cheek, and chin form the perimeter proxy used for ranking.

Presenter-supplied MasqFit screenshots · These images illustrate the workflow, not measurement accuracy.

A preferred perimeter difference, learned from outcomes.

$$\Delta = (P_{face,\ mm} - P_{mask,\ mm}) / 10 \quad [cm]$$

$$\text{Fit score} = \text{sigmoid}[c - \kappa(\Delta - \Delta^*)^2], \quad \kappa > 0$$



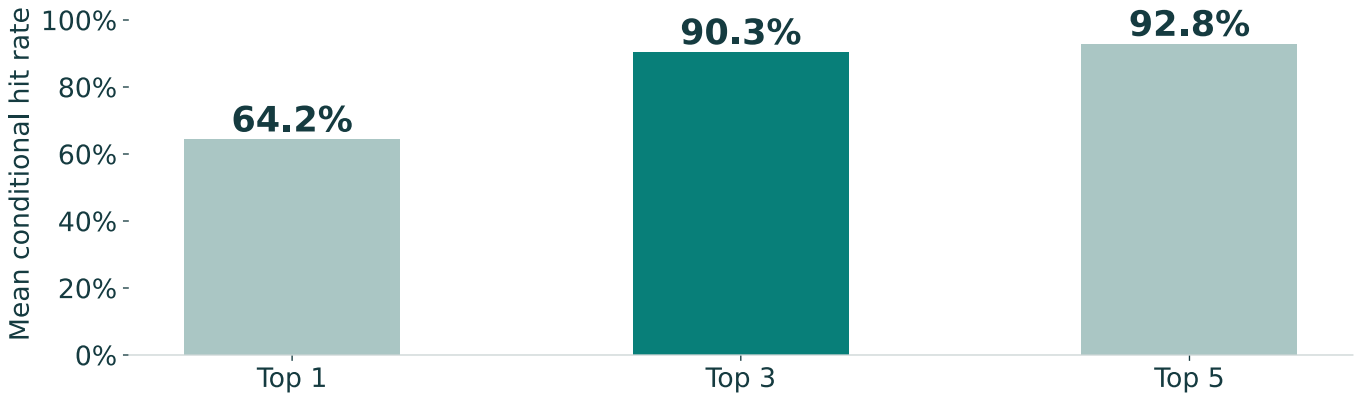
Schematic only · invented coefficients: score = sigmoid[2 − 0.7 × (difference − preferred difference)²]
Not actual mask predictions, fitted coefficients, participant data or validation results.

Δ* is the learned optimum; it need not be zero. Moving away lowers the modeled fit score.
Facial perimeter proxy = nose + upper cheek + mid-cheek + chin paths, measured in mm.

Next feature to test: compare the face's nose path with a matching mask nose perimeter, then evaluate whether it improves held-out fit prediction.

Source: custom_lr calc_preds · Schematic curve uses invented coefficients · Yu & Griffin et al. (2024): FFR-specific face anthropometry and fit.

How often was a fitting result near the top?



Top 1: 64.2% · Top 3: 90.3% · Top 5: 92.8%

Held-out users with ≥1 positive fit label; ranks previously tested observations.

Retrospective relevance evidence. Prospective success with newly recommended masks remains to be tested.

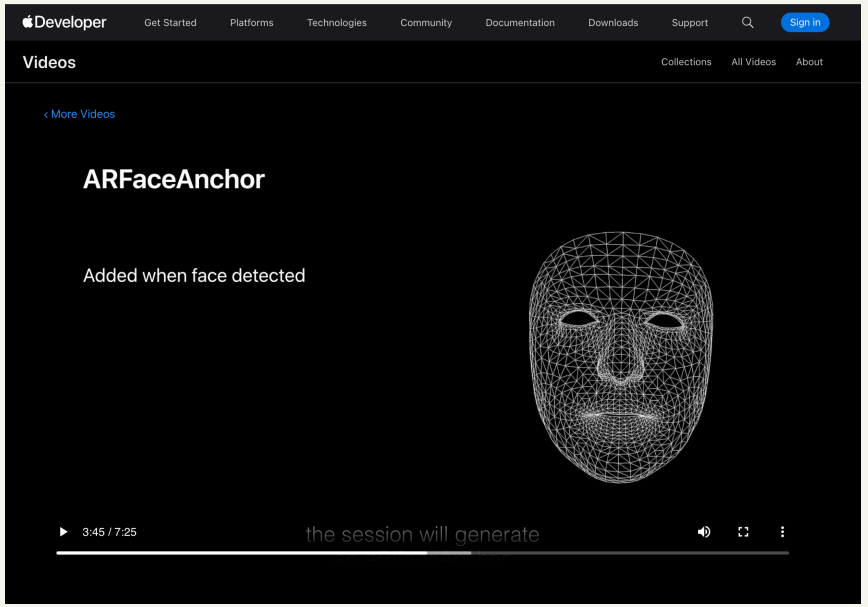
The tradeoff: metric scale, speed, and reach.

Approach	Geometry & scale	Experience	Main constraint
ARKit + TrueDepth Our iOS path	Metric 3D mesh; meters → millimeters	Real-time	Apple hardware path; anatomical mapping still needed
Google MediaPipe	Normalized image coordinates + relative depth; physical scale needs calibration	Real-time	A landmark mesh is not a subject-specific metric measurement
Photogrammetry	Multi-view 3D surface; physical scale must be established	~45 s processing Developer-reported workflow	Slower reconstruction; anatomical landmarking is a separate step

> Standard-camera capture offers a path toward iOS, Android, and webcams.

Apple geometry · Google coordinate conventions · Timing: presenter report, not a controlled comparison.

ARKit made the measurement experience simpler.



Apple Developer: ARFaceAnchor mesh

Metric face geometry

Our TrueDepth path uses infrared depth sensing and a fitted face mesh.

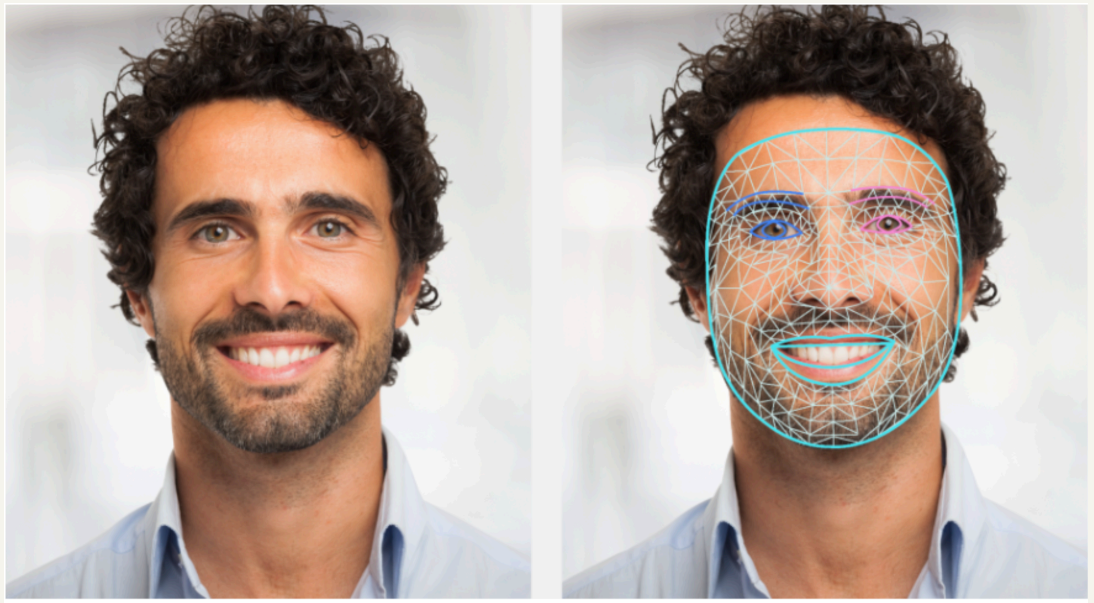
Real-time feedback

No separate pixel-to-millimeter step in this workflow.

Published ARKit study: eye-to-phone distance errors varied from 0.88% to 9.07% across tested conditions. Facial-path accuracy still needs validation.

ARFaceGeometry · ARKit eye-to-phone distance study (2023) · Study did not validate facial-path distances.

A mesh vertex is not an anatomical definition.



MediaPipe example supplied by presenter

Anatomical correspondence

Mesh vertices may not coincide with the traditional landmarks used by NIOSH.

Occluded points

Face models can infer hidden points. An estimated coordinate is not a direct observation.

Open question for ARKit and MediaPipe: how accurate are occluded-point estimates from one viewing angle?

Broader camera support adds new work.

Google: predicted mesh · Apple: fitted face geometry · Single-angle accuracy: not yet established by this project.

Why pursue it

Reconstruct a 3D surface from standard-camera images; a plausible common capture route across platforms.

What it costs

Multi-view capture and a processing wait; motion, coverage, and lighting need attention.

What to add

Physical scale, automatic anatomical landmark detection, and capture-quality checks.

Published facial photogrammetry studies support its potential.
Our camera-only pipeline still needs its own accuracy and repeatability study.

Aldridge et al., 2005: facial photogrammetry vs. direct anthropometry · ~45 s: presenter-reported processing time.

Separate reconstruction from landmarking.

1

3D face

Synthetic geometry
or a captured human face

↓

2

2D landmarks

Detect target points in rendered or captured views

↓

3

3D projection

Map detections onto the matching surface

Validate the handoffs, not only the final score.

Capture → geometry

Measure scale error, repeatability, failed scans, and variation across devices.

↓

Geometry → dimensions

Compare defined landmarks and distances with an independent reference.

Dimensions → fit

Use held-out people and formal fit-test outcomes; evaluate practical selection benefit.

Use one measurement definition across platforms.
Track units, landmark definitions, capture method, and version with every result.

Proposed evaluation priorities; not completed validation results.

Choose the capture route.
Keep the measurement contract.

ARKit reduced calibration friction.

MediaPipe helped with real-time landmarks.

Photogrammetry offers a path to broader camera access.

> Shared research question: which landmarks and reference protocol should both apps use?

Thank you: Kanro · Aaron Collins / OpenAeros · Gerard Hughes · Nukit · anonymous contributors

The camera-only path needs a scale anchor.

Absolute scale

Use an independently known dimension or calibrated capture system.

A canonical average face is a model prior, not an independent ruler for this person.

Surface correspondence

Use the matching camera intrinsics and pose to project a 2D detection onto the mesh.

Check occlusion, multiple intersections, missing surface regions, and agreement across views.

> Evaluate absolute millimeter errors without fitting scale to the reference target.

Sources and boundaries of the claims.

Technical references

Apple: ARFaceGeometry

Apple: face tracking configuration and device support

Google: Face Mesh coordinate systems and canonical model

Photogrammetry evidence

Aldridge et al. (2005), Digital three-dimensional photogrammetry: evaluation of anthropometric precision and accuracy using a Genex 3D camera system

Evidence concerns the tested system and protocol; it does not validate our camera-only pipeline.

Real-time usability, seamlessness, and ~45-second processing are presenter-reported experience. No controlled speed comparison or new measurement-accuracy result is claimed.